02 — Regime Detection

ARCM: Adaptive Rate Curve Model | Liyuan Zhang

Explores the Gaussian HMM regime detection (`regime.py`) and regime-conditional OU calibration (`dynamics.py`).

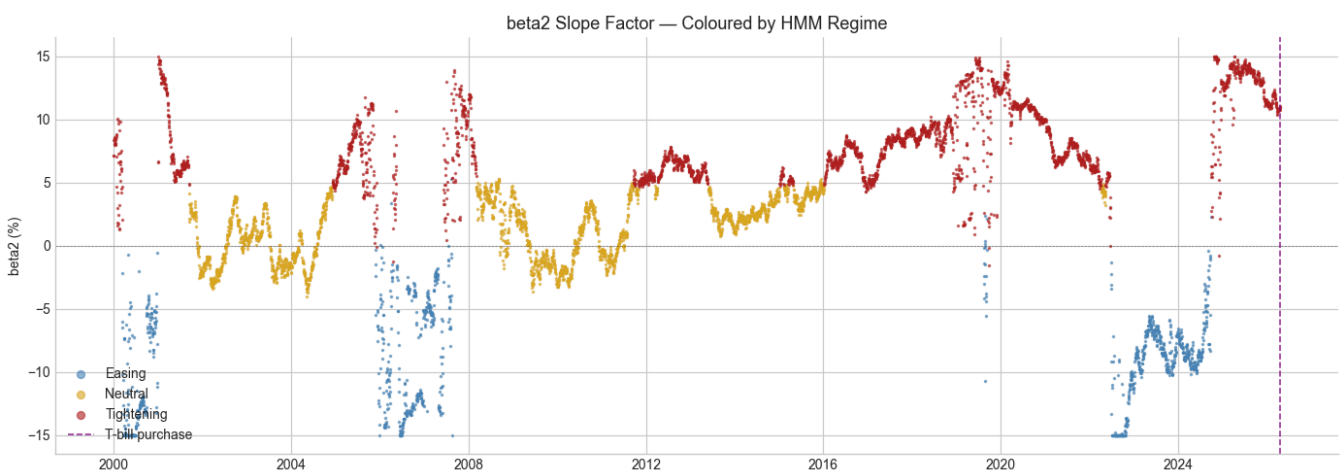
Prerequisite: `pipeline.py` → `regime.py` → `dynamics.py`

1. Load regime outputs

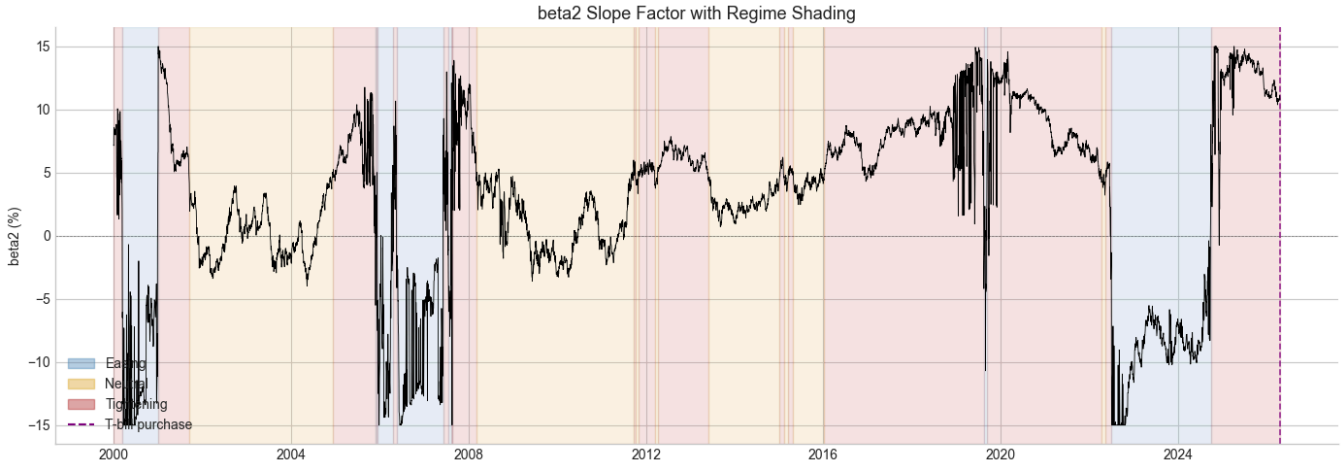
```
Timeline: 6575 rows
Current regime: Tightening  (confidence 100%)
```

	regime_label	n_obs	beta2_mean	beta2_std	avg_episode_days
0	Easing	1161	-8.920305	3.895730	145.125
1	Neutral	2354	1.191264	2.309516	294.250
2	Tightening	3060	8.438556	3.002402	180.000

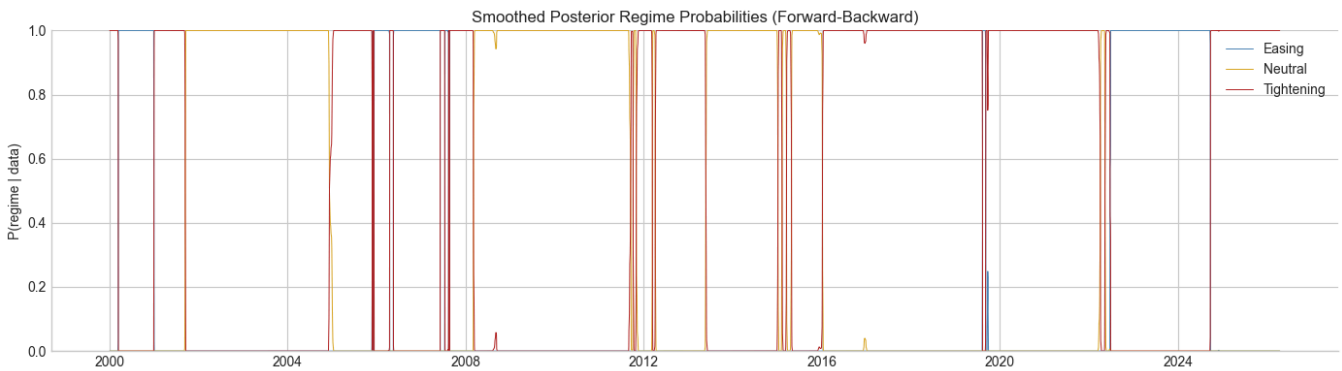
2. Regime timeline — beta2 coloured by detected regime



3. Regime background shading



4. Regime posterior probabilities



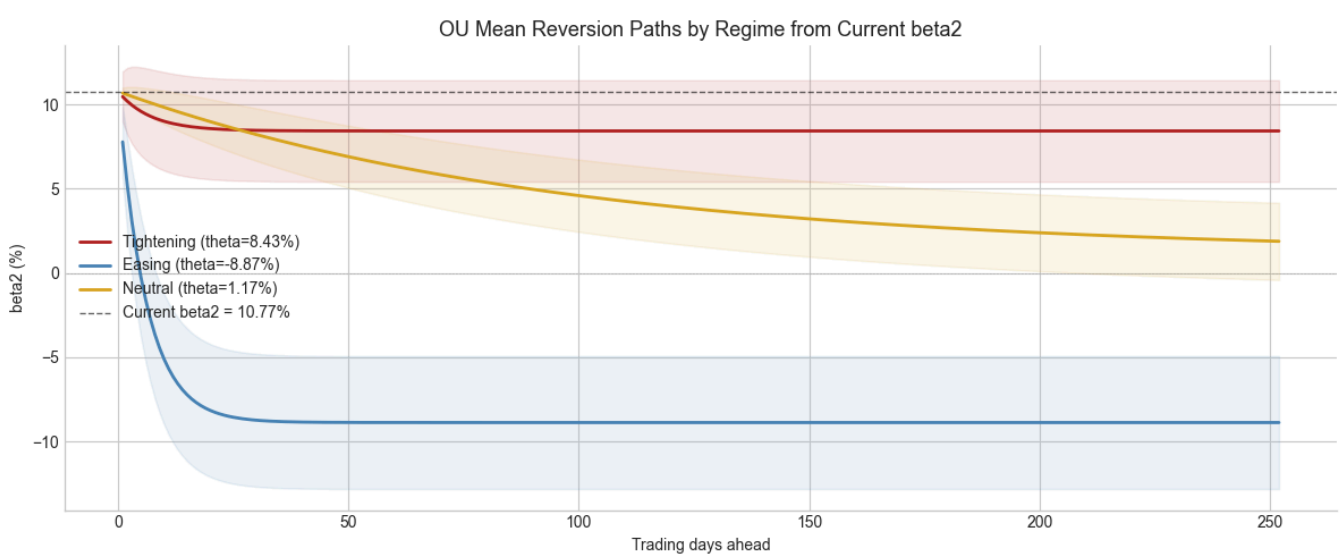
5. OU parameters per regime

Original contribution of ARCM. Each regime has its own mean-reversion speed (kappa), equilibrium (theta), and volatility (eta) — calibrated via closed-form MLE.

```
Regime-Conditional OU Parameters (closed-form MLE)
==================================================================

  Regime        : Tightening
  kappa (speed) : 35.3924  half-life 5 trading days
  theta (equil) : 8.431%
  eta (vol)     : 25.306%
  se(kappa)     : 0.0474
  n obs         : 3044

  Regime        : Easing
  kappa (speed) : 41.6394  half-life 4 trading days
  theta (equil) : -8.868%
  eta (vol)     : 35.956%
  se(kappa)     : 0.1400
  n obs         : 1153

  Regime        : Neutral
  kappa (speed) : 2.5971  half-life 67 trading days
  theta (equil) : 1.169%
  eta (vol)     : 5.234%
  se(kappa)     : 0.0154
  n obs         : 2346
```

6. OU mean reversion paths by regime

Given current beta2, each regime's OU dynamics predict different mean reversion paths over the next 252 trading days (1 year).



7. Position analytics summary

```
Current regime          : Tightening
beta2 current           : +10.774%
OU equilibrium (theta)  : +8.431%
Half-life               : 5 trading days
Expected beta2 @maturity: +8.431%
Expected 1yr yield      : 3.872%
Forecast std            : +/-0.183%
P(rates rise)           : 91.4%
P(rates fall)           : 8.6%
Optimal quantile tau_q* : 0.325
```